# MATH 324 Winter 2025 - Tutorial 6

Section 003 (Thursday)

February 20, 2025

## 1 Method of Moments

Let $\mu_k$ be the $k$-th moment, i.e. $\mathbb{E}[X^k]$. Then $\hat{\mu}_k$ is the $k$-th sample moment,

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} x_i^k.$$

We can write our parameter of interest $\theta$ as a function of the moments, e.g. $\theta = f(\mu_1, \mu_2)$ or $\theta = f(\mu_1)$. Then our method of moments (M.o.M.) estimator is $\hat{\theta} = f(\hat{\mu}_1, \hat{\mu}_2)$, using the sample moments.

Example: $\sigma^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mu_2 - \mu_1^2$. Therefore,

$$\hat{\sigma^2}^{(M.o.M.)} = \hat{\mu}_2 - (\hat{\mu}_1)^2$$

Note: These estimators are often not a function of sufficient statistics. They are generally consistent but often biased.

### Question 1 - Method of Moments

(Exercise 9.73) An urn contains $\theta$ black balls and $N - \theta$ white balls. A sample of $n$ balls is to be selected without replacement. Let $Y$ denote the number of black balls in the sample. Show that $(N/n)Y$ is the method-of-moments estimator of $\theta$.

#### Solution

In this case, $Y \sim$ Hypergeometric and thus $\mathbb{E}[Y] = \frac{n\theta}{N}$ (this can be found in Appendix 2). Thus,

$$\mu_1 = \frac{n\theta}{N} \quad \Rightarrow \quad \theta = \frac{\mu_1 N}{n}$$

In the method of moments estimator, $\hat{\mu}_1 = \bar{Y}$, so

$$\hat{\theta}^{(MoM)} = \frac{\bar{Y}N}{n}.$$

## Question 2 - Method of Moments

(Exercise 9.74) Let $Y_1, Y_2, ..., Y_n$ constitute a random sample from the probability density function given by

$$f(y|\theta) = \begin{cases} \left(\frac{2}{\theta^2}\right)(\theta - y), & 0 \le y \le \theta \\ 0, & \text{elsewhere} \end{cases}$$

(a) Find an estimator for $\theta$ by using the method of moments.

(b) Is this estimator a sufficient statistic for $\theta$?

### Solution

(a)

$$\begin{aligned} \mu_1 = \mathbb{E}[Y] &= \int_0^\theta y \frac{2}{\theta^2}(\theta - y)dy \\ &= \frac{2}{\theta^2} \int_0^\theta \theta y - y^2 dy \\ &= \frac{2}{\theta^2} \left[\frac{\theta y^2}{2} - \frac{y^3}{3}\right]_0^\theta \\ &= \theta - \frac{2\theta}{3} = \frac{\theta}{3} \end{aligned}$$

Therefore, $\mu_1 = \frac{\theta}{3}$ so $\theta = 3\mu_1$ and

$$\hat{\theta}^{(MoM)} = 3\bar{Y}$$

.

(b)

$$L(\theta) = \prod_{i=1}^n f(y_i|\theta) = \prod_{i=1}^n \frac{2}{\theta^2}(\theta - y_i) = \frac{2^n}{\theta^{2n}} \prod_{i=1}^n (\theta - y_i)$$

which cannot be factored into a function that depends only on $\bar{Y}$. Therefore, $\hat{\theta}^{(MoM)}$ is not sufficient for $\theta$.

## Question 3 - Method of Moments

(Exercise 9.76) Let $X_1, X_2, X_3, ...$ be independent Bernoulli random variables such that $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = 0) = 1 - p$ for each $i = 1, 2, 3, ....$ Let the random variable $Y$ denote the number of trials necessary to obtain the first success—that is, the value of $i$ for which $X_i = 1$ first occurs. Then $Y$ has a geometric distribution with $\mathbb{P}(Y = y) = (1 - p)^{y-1}p$, for $y = 1, 2, 3, ....$ Find the method-of-moments estimator of $p$ based on this single observation $Y$ .

### Solution

For a geometric distribution, $\mu_1 = \mathbb{E}[Y] = \frac{1}{p}$. Therefore, $p = \frac{1}{\mu_1}$ and

$$\hat{p}^{(MoM)} = (\bar{Y})^{-1}$$

## 2 Maximum Likelihood Estimation

Objective: Maximize the probability of obtaining the observed sample.

Suppose the likelihood $L$ is a function of $k$ parameters $\theta_1, \theta_2, ..., \theta_k$. Choose as estimators $\hat{\theta}_1, \hat{\theta}_2, ..., \hat{\theta}_k$ to maximize $L(y_1, y_2, ..., y_n|\theta_1, \theta_2, ..., \theta_n)$.

**Invariance property:** If $\hat{\theta}$ is the MLE for $\theta$ and $h$ is a one-to-one function, then $h(\hat{\theta})$ is the MLE for $f(\theta)$.

### Question 4 - Poisson MLE

(Exercise 9.80) Suppose that $Y_1, Y_2, ..., Y_n$ denote a random sample from the Poisson distribution with mean $\lambda$.

(a) Find the MLE $\hat{\lambda}$ for $\lambda$.

(b) Find the expected value and variance of $\hat{\lambda}$.

(c) Show that the estimator of part (a) is consistent for $\lambda$.

(d) What is the MLE for $\mathbb{P}(Y = 0) = e^{-\lambda}$?

#### Solution

(a)

$$L(\lambda) = f(y_1, ...y_n|\lambda) = \prod_{i=1}^{n} f(y_i|\lambda) = \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{y_i}}{y_i!} = e^{-n\lambda} \prod_{i=1}^{n} \frac{\lambda^{y_i}}{y_i!}$$

We want to use the log-likelihood as it's easier to take the derivative of a sum and they share the same max.

$$\begin{aligned}
\ell(\lambda) = \log L(\lambda) &= -n\lambda + \log\left(\prod_{i=1}^{n} \frac{\lambda^{y_i}}{y_i!}\right) \\
&= -n\lambda + \sum_{i=1}^{n} \log(\lambda^{y_i}) - \sum_{i=1}^{n} \log(y_i!) \\
&= -n\lambda + \sum_{i=1}^{n} y_i \log(\lambda) - \sum_{i=1}^{n} \log(y_i!) \\
&= -n\lambda + \log\lambda \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \log(y_i!)
\end{aligned}$$

Take the first derivative to maximize:

$$\frac{d}{d\lambda}\ell(\lambda) = -n + \frac{\sum_{i=1}^{n} y_i}{\lambda} = 0 \quad \Rightarrow \quad \lambda = \frac{\sum_{i=1}^{n} y_i}{n} \quad \Rightarrow \quad \hat{\lambda}^{(ML)} = \bar{Y}$$

(b) $\mathbb{E}[\hat{\lambda}] = \mathbb{E}[\bar{Y}] = \mathbb{E}[Y] = \lambda$
$\text{Var}(\hat{\lambda}) = \text{Var}(\bar{Y}) = \frac{\lambda}{n}$

(c) $\hat{\lambda}$ is unbiased and $\text{Var}(\hat{\lambda}) \to 0$ as $n \to \infty$. Therefore, $\hat{\lambda}$ is consistent for $\lambda$.

(d) By the invariance property, $e^{-\lambda}$ has the MLE $e^{-\hat{\lambda}} = e^{-\bar{Y}}$.

## Question 5 - Gamma MLE

(Exercise 9.84) A certain type of electronic component has a lifetime $Y$ (in hours) with probability density function given by

$$f(y|\theta) = \begin{cases} \left(\frac{1}{\theta^2}\right) y e^{-y/\theta}, & y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

That is, $Y$ has a gamma distribution with parameters $\alpha = 2$ and $\theta$. Let $\hat{\theta}$ denote the MLE of $\theta$. Suppose that three such components, tested independently, had lifetimes of 120, 130, and 128 hours.

(a) Find the MLE of $\theta$.

(b) Find $\mathbb{E}[\hat{\theta}]$ and $\text{Var}(\hat{\theta})$.

(c) Suppose that $\theta$ actually equals 130. Give an approximate bound that you might expect for the error of estimation.

(d) What is the MLE for the variance of $Y$?

### Solution

(a)

$$\begin{aligned} L(\theta|\mathbf{y}) &= \prod_{i=1}^{n} \left(\frac{1}{\theta^2}\right) y_i e^{-y_i/\theta} = \frac{1}{\theta^{2n}} \exp\left\{-\sum_{i=1}^{n} \frac{y_i}{\theta}\right\} \prod_{i=1}^{n} y_i \\ \ell(\theta|\mathbf{y}) &= \log L(\theta|\mathbf{y}) = -2n \log\theta - \sum_{i=1}^{n} \frac{y_i}{\theta} + \log\left(\prod_{i=1}^{n} y_i\right) \\ \frac{d}{d\theta}\ell(\theta|\mathbf{y}) &= \frac{-2n}{\theta} + \frac{\sum_{i=1}^{n} y_i}{\theta^2} \end{aligned}$$

Set $\frac{d}{d\theta}\ell(\theta|\mathbf{y}) = 0$ and we find that $\hat{\theta} = \frac{1}{2}\bar{Y}$.

From the observations, $\bar{y} = \frac{1}{3}(120 + 130 + 128) = 126 \Rightarrow \hat{\theta} = 63$.

(b) Recall that for $X \sim \text{Gamma}(\alpha, \beta)$, $\mathbb{E}[X] = \alpha\beta$ and $\text{Var}(X) = \alpha\beta^2$. Thus,

$\mathbb{E}[\hat{\theta}] = \mathbb{E}\left[\frac{1}{2}\bar{Y}\right] = \frac{1}{2}\mathbb{E}[Y] = \frac{1}{2}2\theta = \theta$

$\text{Var}(\hat{\theta}) = \frac{1}{4}\text{Var}(\bar{Y}) = \frac{1}{4}\frac{\text{Var}(Y)}{n} = \frac{1}{4}\frac{2\theta^2}{3} = \frac{\theta^2}{6}$.

(c) As we have very low $n$, we will use the 2 standard deviation bound:

$$2\sqrt{\text{Var}(\hat{\theta})} = 2\sqrt{\frac{130^2}{6}} = 106.14.$$

(d) $\text{Var}(Y) = 2\theta^2$. By the invariance property of the MLE, $\widehat{\text{Var}(Y)} = 2(\hat{\theta})^2$.

## Question 6 - MLE for proportions

(Exercise 9.87) A random sample of 100 voters selected from a large population revealed 30 favoring candidate A, 38 favoring candidate B, and 32 favoring candidate C. Find MLEs for the proportions of voters in the population favoring candidates A, B, and C, respectively. Estimate the difference between the fractions favoring A and B.

### Solution

Let $Y_1$ be the number of voters favouring candidate A, $Y_2$ be the number favouring B and $Y_3$ the number favouring C. Then $(Y_1, Y_2, Y_3) \sim \text{Multinomial}(p_1, p_2, p_3)$.

In the likelihood, we only need to consider $(p_1, p_2)$ since $p_3 = 1 - p_1 - p_2$.

$$L(p_1, p_2|\mathbf{y}) = \frac{n!}{y_1!y_2!y_3!} p_1^{y_1} p_2^{y_2} (1 - p_1 - p_2)^{y_3}$$

We jointly maximize w.r.t. $p_1$ and $p_2$ to get $\hat{p}_1 = Y_1/n = 0.30$, $\hat{p}_2 = Y_2/n = 0.38$, and $\hat{p}_3 = Y_3/n$. Then, the MLE of $p_1 - p_2 = \hat{p}_1 - \hat{p}_2 = -0.08$.

## Question 6* - MLE for continuous uniform

(*This was an extra question that was covered.*) Let $X_1, X_2, ..., X_n$ be i.i.d. with p.d.f.

$$f(x|\theta) = \frac{1}{\theta}, \quad 0 \le x \le \theta, \quad \theta > 0.$$

Estimate $\theta$ using both the method of moments and maximum likelihood. Calculate the means and variances of the two estimators. Which one should be preferred and why?

### Solution

Method of Moments: $\mu_1 = \mathbb{E}[X_i] = \frac{\theta}{2} \quad \Rightarrow \quad \theta = 2\mu_1 \quad \Rightarrow \quad \hat{\theta}^{(MoM)} = 2\bar{X}$.

As $X \sim \text{Unif}(0, \theta)$, $\mathbb{E}[\hat{\theta}^{(MoM)}] = \mathbb{E}[2\bar{X}] = 2\mu_1 = \theta$. Meanwhile, $\text{Var}(X_i) = \theta^2/12$ so $\text{Var}(\hat{\theta}^{(MoM)}) = 4\text{Var}(\bar{X}) = \frac{\theta^2}{3n} \approx O(1/n)$.

Maximum likelihood: We can write the p.d.f. using the indicator as

$$f(x|\theta) = \frac{1}{\theta} I_{(0,\theta)}(x).$$

Then,

$$L(\theta|\mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\theta} I_{(0,\theta)}(x_i) = \frac{1}{\theta^n} \prod_{i=1}^{n} I_{(0,\theta)}(x_i) = \frac{1}{\theta^n} I_{(0,\theta)}(x_{(n)})$$

(as the product is only nonzero if $x_{(n)} \le \theta$).

For $\theta \ge x_{(n)} = \max(x_1, ..., x_n)$, $L(\theta|\mathbf{x}) = \frac{1}{\theta^n}$ which is decreasing in $\theta$. For $\theta < x_{(n)}$, $L(\theta|\mathbf{x}) = 0$. Therefore the max of $L(\theta|\mathbf{x})$ is at $x_{(n)}$, so $\hat{\theta} = X_{(n)}$.

(*For more details about order statistics, refer to Section 6.7 in the textbook.*) The p.d.f. of $X_{(n)}$ is $\frac{nx^{n-1}}{\theta^n}$ for $0 \le x \le \theta$. This means $\mathbb{E}[\hat{\theta}] = \mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta$ and $\text{Var}(\hat{\theta}) = \frac{n\theta^2}{(n+2)(n+1)^2} \approx O(1/n^2)$.

We have that $\hat{\theta}^{(ML)}$ is biased, but this bias is small if $n$ is large. $\hat{\theta}^{(MoM)}$ is unbiased. $\hat{\theta}^{(ML)}$ has a smaller variance that $\hat{\theta}^{(MoM)}$.